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PROJECT BRIEF

Mobile Development
PROG421-EC10

Project Title: CampusGo – Antonine Campus Delivery Application

1. Introduction

On a busy university campus, students and staff often have only a few minutes between two classes or lab sessions. They cannot always leave their classroom or lab to buy a sandwich, a coffee, a notebook, or to print a document before the next session. This project aims to build a **mobile-only campus delivery application** using **React Native** that allows the students and staff of **Université Antonine** to order from the on-campus services (cafeteria, coffee corner, bookstore & stationery, and printing office) and to receive their orders **directly in a specific building and room**, **within a chosen class break**.

2. Aim

The primary goal of this project is to develop a cross-platform mobile application that allows users to **browse campus services, place orders, print documents, and receive them in their classroom** at the right time. The application should focus on **user experience**, **speed**, and **reliability of the business rules** (opening hours, delivery slots, wallet balance), so that an order never arrives after the break is over.

3. Objectives

- Develop a **React Native mobile application** that allows users to:
  - Sign in with a university account and manage their profile
  - Browse campus stores and products, with live opening hours
  - Order printing jobs with automatic price calculation
  - Receive orders in a specific building and room, within a selected class break
  - Pay with a campus wallet and track the order status
  - View the order history and reorder previous orders
- Implement a simple and intuitive user interface using tab and stack navigation
- Apply data validation and business rules (opening hours, minimum order, delivery slots, wallet balance)

4. Requirements and Constraints

React Native Mobile Application

General Technical Requirements

- The application must be built with **React Native** (Expo is allowed) and **React Navigation** (bottom tabs + stack navigator).
- Use functional components and Hooks (**useState**, **useEffect**); all lists must be displayed with **FlatList**.

In this document, the masculine form refers, as appropriate, to both women and men. The use of the masculine is for the sole purpose of facilitating the reading of the text and has no discriminatory intent.

- Data (stores, products, buildings, delivery slots) must come from a local JSON file or a mock API (e.g., json-server). The cart, the wallet and the orders must be saved with **AsyncStorage**.
- No real backend, real payment or real GPS tracking is required: payments and order tracking are **simulated**.
- The application must run on a real phone (Expo Go) or an emulator during the demonstration.

Screen 1: Welcome / Login
Purpose: Authenticate the students and staff of the university.
Features:

- App logo, name and a short introduction message
- Login form with a university email (must end with **@ua.edu.lb**) and password
- Sign-up form: full name, student/staff ID, faculty, phone number, password
- Form validation (email domain, ID format, password length) with clear error messages

Screen 2: Home – Campus Stores
Purpose: Central hub showing all the campus services.
Features:

- List of campus stores (Cafeteria, Coffee Corner, Bookstore & Stationery, Printing Office) with logo, category and average preparation time
- **Open / Closed** badge computed from each store's opening hours and the current time; closed stores cannot receive orders
- Search bar and filter by category
- Banner showing the next available delivery slot and the current wallet balance

Screen 3: Store Menu & Product Details
Purpose: Allow users to browse the products of a store and add them to the cart.
Features:

- Products grouped by category (e.g., Sandwiches, Hot Drinks, Notebooks) with image and price
- Product details with options (size, extras) and a quantity selector
- Out-of-stock products are displayed but cannot be added
- Cart badge showing the number of items
- Rule: the cart can contain products from **one store only**; adding a product from another store asks the user to empty the cart first

Screen 4: Printing Service
Purpose: Allow users to order printed documents delivered to their classroom.
Features:

- Form: document name (simulated upload), number of pages, number of copies, color or black & white, single or double-sided, binding (none / staple / spiral)

- **Live price calculation** from a price table (e.g., B&W $0.05/page, color $0.25/page, spiral binding +$1.50), updated each time an option changes
- Validation: pages and copies greater than 0, maximum 200 pages per order
- Add the printing job to the cart of the Printing Office

Screen 5: Cart & Checkout
Purpose: Review the order, choose where and when to receive it, and pay.
Features:

- List of items with quantity (+ / −), remove button, subtotal, delivery fee and total
- Delivery fee rules: free delivery above $10, otherwise $0.50; minimum order $2
- Delivery location: choose the **building** (from a list) and enter the **room number** (required)
- Delivery time: choose a **class break slot** (e.g., 9:45–10:00, 11:15–11:30, 12:45–13:30); slots that are already passed or too close to the store's preparation time are disabled
- Payment method: Campus Wallet or cash on delivery; wallet payment is refused if the balance is insufficient
- Order summary and confirmation

Screen 6: Order Tracking
Purpose: Follow the order until it reaches the classroom.
Features:

- Status timeline: Received → Preparing → On the way → Delivered to Building X, Room Y (simulated: the status changes automatically every few seconds using a timer in useEffect with cleanup)
- Courier name and estimated delivery time
- Cancel button available only while the order is still "Received"
- Rate the order (1 to 5 stars + comment) after delivery

Screen 7: Orders History, Profile & Wallet
Purpose: Manage past orders, personal information and the campus wallet.
Features:

- List of previous orders with date, store, total and status; filter by store or status
- **Reorder** button that copies a previous order into the cart
- Profile: name, ID, faculty, and saved delivery locations (e.g., "My Lab – Building B, Room 204")
- Wallet: current balance, top-up (simulated) and list of wallet transactions

5. Deliverables

- Fully functional **React Native mobile application**
- Presentation and live demonstration showcasing core features and user experience
- Codebase and project documentation (GitHub or ZIP)
- Mock data file (JSON) with at least **4 stores**, **25 products**, **6 buildings** and **5 delivery slots**

- User guide or instruction manual

6. Evaluation Rubric

The project will be assessed based on the following criteria.

Criteria (Based on the CLO/Kpi)	Beginning 1	Progressing 2	Developing 3	Exemplary 4	Coef
Understanding of Concepts	Limited understanding of basic concepts and principles.	Partial understanding of basic concepts and principles.	Demonstrates a solid understanding of basic concepts and principles.	Demonstrates a deep understanding of advanced concepts and principles.	20
Front End Design	Unable to formulate a clear and effective design approach for front-end development tasks.	Partially formulates a design approach but lacks coherence or effectiveness in translating concepts into design elements.	Formulates a design approach for front-end development tasks with some effectiveness, demonstrating understanding of design principles.	Formulates a well-defined and effective design approach for front-end development tasks, demonstrating proficiency in design principles and implementation techniques.	20
Presentation Skills	Unable to deliver a coherent presentation, lacking organization and clarity in conveying ideas.	Partially presents ideas but struggles with maintaining audience engagement and coherence in delivery.	Presents ideas with some effectiveness, demonstrating organization and clarity in communication, but may lack engagement.	Delivers a well-structured and engaging presentation, effectively conveying ideas with clarity, coherence, and audience engagement.	15
Overall Functionality	Demonstrates inadequate functionality, with numerous bugs, errors, or missing features affecting usability.	Partially achieves functionality goals but may have significant issues or limitations that impact overall usability and	Achieves functionality goals with some effectiveness, providing a satisfactory level of usability and user experience, but	Achieves overall functionality goals effectively, delivering a high level of usability and user experience with minimal	45

		user experience.	may still have minor issues or room for improvement.	issues or limitations.	

7. Project Timeline and Follow-up Schedule

All deliverables are graded.

- Project Kickoff: __/__/2026
- Presentation and Evaluation: last 2 sessions

8. Resources and Support

- Access to well-equipped computer labs and software tools for development and testing purposes, ensuring that team members have the necessary resources to work efficiently.
- Availability of relevant textbooks, research papers, and online resources on React JS, React Native and other relevant topics to support learning and problem-solving.
- Guidance and support from the project supervisor or instructor, who will be available during designated office hours or via email to provide assistance, answer questions, and offer feedback on project progress.
- Regular progress meetings or checkpoints to review project milestones, address any issues or concerns, and ensure that the project is on track for successful completion.

Please visit the following references:

1. React Documentation

- React **Native Documentation**

Website: https://reactnative.dev
Description: Official documentation for React Native including setup, components, APIs, and tutorials.

- React **Navigation**

Website: https://reactnavigation.org
For implementing stack, tab, and drawer navigation in React Native.